

# Ryan Qin

☎ (416)-939-4721 | ✉ r25qin@uwaterloo.ca | [🌐 LinkedIn](#) | [🐙 Github](#) | [🌐 ryanqin.com](#)

## Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

## Technical Skills

**Languages:** C, C++, C#, Assembly, Verilog, Java, Python, HTML/CSS, JavaScript

**Technologies:** Robotics, RTOS, Linux, STM32, DSP, FSM, FPGA, TCP/IP, MongoDB, Node.js, React

**Development Tools:** TensorFlow, PyTorch, Android, AWS, Git, Cadence, Altium, MATLAB, DSO, Fusion 360

## Professional Experiences

**I/O Compliance Engineer** | TensorFlow, OpenCL, FPGA, C++, C#, Cadence, DSO,  $I^2C$   
AMD

May 2026 - Present

- Designing **FPGA-based** USB 2.0 I/O testing board, aimed to **autonomously** test Link Layer and drop/droop compliance
- Developing **STM32-based**  $I^2C$  module for **automatic** HDMI **EDID** setting, helping to reduce manual rewiring work
- Optimizing electrical Compliance analysis calculations using **OpenCL**, utilizing GPU parallelism for consecutive test runs
- Implementing **USB 4.0 test-automation** software in **C#**, designed to automate instruments over **TCP/IP** protocol

**Electrical Engineer** | C++, OpenCV, TCP/IP, LtSpice, Altium  
ATS Corporation

Sep 2025 - Dec 2025

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Developed an **inductively isolated** pulse conditioner board in **Altium**, protecting PLC against stray voltage while **reducing 80%** of **Sub-GHz noise** from turbocharger's hall-effect tachometer
- Designed and implemented **PID control loops** and **dither circuits** for a fuel nozzle testing system's pressure regulator, enabling **25%** more accurate simulation of turbojet engine operating conditions.
- Implemented **OpenCV** jet spray analysis, reducing test time by **30%** through automated spray angle and stability measurements

**Electrical Coordinator** | C#, .Net, MySQL, COMSOL, Python  
WSP

Jan 2025 - May 2025

- Recognized by **WSP Canada's** Rail and Transit **VP** for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (**OCS**) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by **10 years**
- Implemented AI-driven weather prediction using **Prophet**, improving storm and temperature tolerance by **30%**

## Projects

**FPGA Clock** 🚩 | Verilog, Vivado, FSM, FPGA

Dec 2025 - Jan 2026

- Developed a **real-time** VGA digital clock for FPGA using **Verilog**, designing **Moore FSM** for time counters, VGA timing, and character rendering using Vivado and HDL simulation
- Implemented **parameterized RTL** to simplify character resizing and repositioning across display resolutions

**Autodump** 🚩 | I/O system Design, C++, TCP/IP, SSH, Python, OpenCV

Nov 2025

- Developed an **autonomous** self-emptying trash robot, achieving an **overall 3<sup>rd</sup>** place at **MakeCU**
- Engineered an Arduino-based serial **distributed I/O** system, focused on **simplifying** software-hardware integration and **reducing PWM-signal jitter** and **integrity** for servo and ultrasonic sensors
- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously navigate** back to home position

**Competition RC** 🚩 | PyTorch, FreeRTOS, LTSpice, AVR Assembly

May 2025

- Built an RC car with **50%** recycled material; selected as the **top qualified finalist** out of 15 teams
- Designed a **custom** airsoft motor controller using MOSFET stages, relays, and Arduino for PWM signal conversion, achieving a **fast lap record** of **17 seconds** compared to **average 25 seconds** lap time
- Implemented a **lane-keeping algorithm**, combining **PID** and **PyTorch**-based vision to adjust steering from USB camera

**Baymax Bot** 🚩 | Machine Learning, Python, C++, STM32, TCP/IP

Jan 2025

- Trained a voice model using **Applio**, replicating BayMax's Voice with **customized response** from **ChatGPT's API**
- Implemented a UART-based **STM32 field I/O** controller to **offload** time-critical motor coordination from a **Raspberry Pi**